Assignment Answers

1. A database is an organised collection of data that is stored electronically so that it can be accessed, managed and updated
- Databases are essential because they help businesses/organisations store large amounts of data efficiently and securely. Through this we can quickly search, report and make decisions when needed.

2 real world examples include: - Banks using databases to store customer accounts and transaction details
- Hospitals use databases to store patient records and appointments.

2. A datawarehouse is a system used to store structured data in large amounts for fast processing and analysis
- A data warehouse

3. A data lake is a system that keeps/stores raw structured, semi-structured and unstructured data in it's original format
- A data lake differs from a data warehouse because a warehouse stores clean and organised data while a data lake stores raw data.

- Businesses prefer data lakes when dealing with large volumes of different data types such as posts, emails, logs, and sensor data.

4. A data mart is a smaller version of a data warehouse, it is focused on a specific department such as finance, sales or marketing.
- A data Mart gives teams quick access to relevant data without needing the full data warehouse.

for simple sharing
good for spreadsheets -

used for streaming data)

allig require human intelligence

of bolg following programmed

es

with many layers
They process inputs through

age so that it can respond

stand and generate language
at a very large scale

- A data mart could commonly be used by managers, analysts, and department staff.

5. Structured data: Organised in rows and columns. e.g Excel spreadsheet
Semi-structured data: Has some organisation but not fixed tables e.g JSON file
Unstructured data: No fixed format E.g Images, videos, emails

6. Data modelling is a process of designing how data will be stored, collected and managed by a system
- It is important because it improves efficiency, accuracy and scalability

Types of data models: - Conceptual Model: High level overview of business data
- logical Model: Detailed structure without technical details
- physical: actual database design.

7. Data mining is the process of finding patterns, trends and useful insights from large datasets
2 examples: - Retail stores analysing customer buying habits
- Banks detecting fraud by spotting unusual transactions

8. A DBMS is a software used to create, manage and control databases
Examples: - MySQL: Advanced open-source database for complex systems
- PostgreSQL: Advanced open-source database for complex systems
- Oracle: Enterprise database used by large organisations
- MongoDB: NoSQL database used for flexible data like apps

9. ETL (Extract: collect data from different sources, Transform: Clean and prepare data, Load: Store the data into a warehouse or system)

ETL: combines data into one usable source for analysis.

10. CSV: Stores table data using a comma as a delimeter (used for simple sharing) good for spreadsheets.

JSON: flexible format used for APIs.

parquet: fast compressed format for analytics

Avro: Row-based format used in big data systems (used for streaming data)

11. AI is the ability of machines to perform tasks that normally require human intelligence

Narrow AI: Designed for one task e.g Siri

General AI: Human-like intelligence accross many tasks.

12. ML is a type of AI where systems learn from data instead of only following programmed rules.

Traditional programming: Rules + data = answers

Machine Learning: Data + Answers = learns rules

Types: Supervised learning: Uses labelled data

Unsupervised " : finds hidden patterns

Reinforcement " : Learns through rewards and penalties

13. Deep learning is a branch of ML that uses neural networks with many layers

- Neural networks are systems inspired by the human brain. They process inputs through connected nodes to recognise patterns.

14. NLP allows computers to understand and interprete human language so that it can respond

e.g: google translate, Chatgpt + Chatbots; Email Spam filters

15. An LLM is a model trained on huge amounts of text to understand and generate language

- It differs from ML models because it handles language tasks at a very large scale

E.g ChatGpt: Conversation, writing, coding

- Claude: research, writing, reasoning

16. Generative AI creates new content such as text, images, music or code
- At a high level it works by learning patterns from training data and generating similar outputs
e.g Chatgpt: generates text
DALL-E: Creates images
Github copilot: Writes code

17. A training dataset is data used to train an AI model
- quality is important because poor data = poo results, ~~ab~~ biased data would lead to unfair / skewed results or decisions

18. Computer vision allows machines to analyse and understand images or videos
- Machines use algorithms and neural networks to detect shapes, objects and patterns
e.g: face recognition
- Self-driving cars recognising roads and pedestrians

Section C

19. An API is a tool that allows two systems to communicate
- API's are important because they connect apps and share data automatically
- API use: A waiter in a restaurant takes your order to the kitchen and brings back food. The waiter in a API between you and the kitchen

20. An API key is a secret code used to identify and authorise access to an API
- API keys should be stored securely in an environment variables or private files
- If accessed publicly, others may misuse the service or create charges

21. REST API stands for representational State Transfer

methods: GET - Retrieve data

- POST - Create data
- PUT - update "
- DELETE - Remove "

22. A JSON is a lightweight data format used to exchange data.
- It is commonly used because it is easy for humans and machines to ~~use~~ read.

E.g {
"name": "John",
"age": 25,
"city": "Johannesburg."
}

23. prompt engineering is writing effective instructions for AI tools
E.g poor prompt: Write about business
Better " : Write a 200 word explanation of how small businesses implement data analytics

poor prompt: Help me code
Better " : Write a python program that sorts a list using bubble sort

24. A map function applies the same action to every item in a list
E.g numbers = [1,2,3]
result = list(map(lambda x: x * 2 numbers))
Output = 2, 4, 6

25 Cloud Computing means using internet based computing resources
models: Iaas - Infrastructure (AWS, E.
Paas - Platform (Google App Engine)
Saas - Software (Microsoft 365)

26. A version control system tracks file changes over time.
Git is the most popular version control
- Commit: Saved version of work
- Branch: Separate copy of new changes

27. Jupyter notebook is an interactive tool for writing code, notes and visuals in one place
- It is commonly used ~~for~~ with python
- popular because analysts can run code step-by-step and explain results

28 python is a high level programming language known to be simple and easy
- popular for analytics and AI because it is easy to learn and has many libraries
Libraries: pandas - Data Analysis
Numpy - Numerical Computing
Matplotlib : Charts
~~Sk~~ Scikit-learn: Machine learning

Section D

29. Data privacy is protecting personal information and controlling how it is used
- GDPR is a European union law that protects personal data of EU residents

30. Data ethics means using data fairly and responsibly.
Examples of unethical use: - Selling user data without consent
- Biased hiring AI
- Misleading deepfake values

31 Data governance is managing data quality, security, ownership and policies
Key components: Data quality - compliance - Standards and policies
- Security - Roles and responsibilities

32. BIAS in AI means unfair outcomes caused by biased data or design
e.g facial recognition working poorly on darker skin tones

33

3.

34 BI uses tools and data to help businesses make better decisions
Tools: Power BI, Tableau
These create dashboards, reports and trends

Section E

35 The most interesting concept would be AI because it can learn patterns and automate tasks that normally need human thinking.

36. The most complex would be Machine learning because neural networks have many layers and can be complex. I used youtube tutorials, google articles and examples to try and understand

37 Data and AI will affect my future career by creating more opportunities in analytics and decision making. E.g businesses use dashboards to track peformance and AI tools to improve customer service. In reality AI helps with maps, recommendations, and smart assistants

References

1. IBM
2 Google Cloud
3. Microsoft azure
4. Chatgpt
5. AWS
6. postgreSQL
7. MongoDB
8. EU (GDPR offical website)

26. A version control system t
Git is the most popular v
- Commit: Saved version
- Branch: Seperate copy

27. Jupyter notebook is
one place
- It is commonly u
- popular because anal

28 python is a high lev
- Popular for analytic
Libraries: pand
Nump
Matpl

Section D

29. Data privacy is
- GDPR is a E

30. Data ethics n
Examples of

31 Data governa
key component